Dania Orfali

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EXPERIENCE

Lawrence Berkeley National Lab (LBNL) — **Jan 2026 – Present**
Research Doctoral Fellow — California, USA

- Designed and fabricated a novel gas-Membrane Electrode Assembly (MEA) cell compatible with operando X-ray spectroscopy
- Conducted synchrotron X-ray absorption spectroscopy (XAS) fluorescence experiments
- Performed and analysed Cu K-edge XANES and EXAFS measurements of to resolve catalyst oxidation state and coordination environment under millisecond scale pulsing electrochemistry, with inline product characterization (GC, MS & NMR)
- Sputtered Cu catalyst films to fabricate gas diffusion electrodes (GDEs)

KAIO Labs, HAX/SOSV — **June 2025 – August 2025**
Ph.D. Research Intern — New Jersey, USA

- Designed and executed CO_2 electroreduction experiments under industrially relevant conditions (>300 mA cm^{-2})
- Spray-coated composite electrodes using metal-based catalyst inks (Au, Ni, Cu, Ag, Co)
- Systematically screened catalyst compositions (metals, solvents, ionomers) and operating conditions (temperature, pressure, flow rate, electrolyte/ionomer ratio) to map performance space
- Performed electrochemical impedance spectroscopy (EIS) and characterized electrode behavior via GC and SEM/EDS
- Achieved electrodes exceeding literature and commercial benchmarks (FE >90%) using ML-guided screening

Novo Nordisk Foundation Center for Biosustainability (DTU Biosustain) — **Feb. 2023 – July 2023**
Biological and Chemical Engineer Researcher, Intern — Copenhagen, Denmark

- Designed the full process of sugars extraction from macroalgae for industrial feedstock use using Aspen HYSYS/Aspen Plus
- Performed a technoeconomic analysis (TEA) and life cycle assessment (LCA) of sugar extraction process from macroalgae
- Quantified and ranked the economic and environmental performances of all conventional and novel biobased feedstocks for biotechnology using a standardized method

Norwegian Institute of Bioeconomy Research (NIBIO) — **May 2022 – June 2022**
Wood Technology Researcher, Intern — As, Norway

- Valorized treated wood waste through fabrication of particleboards for material reuse applications
- Evaluated dimensional stability (swelling), mechanical properties, and surface wettability (contact angle) of produced boards

EDUCATION

Ph.D., Chemical and Biomolecular Engineering — **Sep. 2023 – Present**
New York University, New York, USA — GPA: 4.0/4.0
Advisor: [Miguel A. Modestino]
Research Interests: CO_2 reduction, pulsed electrochemistry, catalyst activity, Bayesian Optimization & ML, experimental automation

Masters, Biological and Chemical Engineering *(Erasmus Mundus Scholarship)* — **Sep. 2021 – Aug. 2023**
Technical University of Denmark, AgroParisTech (Université Paris-Saclay), Tallinn University of Technology & Universite de Reims Champagne Ardenne - Denmark, France & Estonia — GPA: 3.95/4.0
Research Interests: Biomaterials, bioprocessing, biotechnology, circular bioeconomy, TEA, LCA

Bachelor of Engineering, Chemical Engineering — **Sep. 2017 – May 2021**
American University of Beirut, Beirut, Lebanon — GPA: 3.86/4.0
Courses: Numerical methods, thermodynamics, fluid dynamics, transport phenomena, process control, reactor engineering

PROJECTS

Operando Cu K-Edge XAFS of pulsed CO_2 electroreduction in a high-current MEA: Linking second and sub-second catalyst dynamics to product selectivity

- Conducted high-throughput in-situ synchrotron XAS fluorescence experiments on an MEA cell to resolve catalyst oxidation state and coordination environment under pulsing electrochemistry
- Fabricated a novel MEA cell compatible with operando X-ray measurements operating at current densities >300 mA cm^{-2} and C2+ selectivity >50%

Automated optimization of dilute CO_2 electrolyzers using pulsing and active learning algorithms

- Implemented electrochemical pulsing protocols for dilute industrial CO_2 streams, improving product selectivity by 207% and energy productivity by 241% relative to steady-state operation
- Built an automated MEA CO_2 electrolyzer platform integrating LabVIEW, MATLAB, and ML algorithms to identify optimal conditions across varying feed compositions, with real-time adaptation to process disturbances
- Developed adaptive learning algorithms for rapid optimization of Cu-based multiproduct CO_2 electrolyzers under dynamic, renewable-relevant power profiles, maintaining <20% product variability across 20–100% power fluctuations

Lignin engineering for the production of a multifunctional additive for plastics

- Chemically modified Organosolv lignin via chloromethylation and esterification, confirming functionalization by FTIR with up to 92.5% mass recovery
- Fabricated PLA/modified lignin composites, showing improved processing instabilities and thermal degradation behavior
- Characterized pine and barley straw lignin via SEC, FTIR, and 2D-HSQC NMR, identifying as superior feedstocks to aspen wood

L'Oreal Brandstorm Competition 2020: Building a plastic-less future in the beauty industry *(Reached Top 3 at country level)*

- Pioneered a sustainable bioplastic packaging solution from coffee waste for L'Oréal, validated through LCA
- Performed techno-economic and market analysis to assess production feasibility and consumer demand

PUBLICATIONS

Pulsed Electrolysis Promotes Catalyst Activity in Dilute CO_2 Streams
ACS Energy Letters, 2026

Ranking economic and environmental performance of feedstocks used in bio-based production systems
Current Research in Biotechnology, 2025 | Also a poster presentation at Metabolic Engineering Conference, 2024

Reimagining Unit Operations Labs: NYU's Multi-Tiered Experiential Learning Platform.
AIChE Annual Meeting, 2025

PATENTS

METHODS OF PULSED ELECTROLYSIS FOR REDUCTION OF CARBON DIOXIDE FROM MIXED GAS STREAMS
Application No. 63/996,089 and approved filing receipt by USPTO on April 22, 2026

AWARDS, FELLOWSHIPS, AND CERTIFICATIONS

Advanced Light Source Doctoral Fellowship
Granted by Lawrence Berkeley National Lab 2026 **-** UC Berkeley

School of Engineering (SoE) Fellowship
Granted by New York University for a Ph.D. in Chemical Engineering for the year of 2023-2024

Postgraduate program in Data Science and Business Analytics *(Online)*
Granted by Purdue University. Program from June 2022 – Jan 2023, Certificate ID: 65736575

SKILLS

- GC, MS, NMR, FTIR, UV-Vis, X-ray spectroscopy, SEM/EDS, spray coating, sputtering, CO_2 reduction, chromatography, PCR, gel electrophoresis, Western blot, RNA/DNA/lipids extraction, cell culture, genetic engineering
- LabVIEW, Python, Matlab, R, COMSOL, Simulink, ML-based optimization and automation, AutoCAD, LCA, TEA, Aspen HYSYS

MISC.

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Legal Name: Dania Muhieddine Orfali
Available for internship: June 2027